

MICROVISION

Service Management System

General Requirements Document (GRD) — Version 1.1

LED | Coolers · Rajasthan & Punjab Operations

1. Project Overview

Microvision is a company dealing in LED lights and coolers, operating across 50+ cities in Rajasthan and parts of Punjab. The company manages installation and complaint service requests through a network of 400–600 city-based service centres.

Currently, complaints received via phone calls are manually routed to service centres, creating inefficiencies in tracking, billing, and communication. This system replaces that manual process with a centralized web-based platform.

1.1 Purpose

- Provide a single platform for registering, assigning, tracking, and resolving service complaints.
- Enable service centres to receive, accept, and update complaint status digitally.
- Automate billing calculations and generate monthly invoices per service centre.
- Notify service centres via WhatsApp on new assignments and reopens.
- Give admin full visibility across all cities, centres, complaints, and payments.

1.2 Business Context

Parameter	Detail
Product lines	LED lights, Coolers
Operating regions	Rajasthan (primary), Punjab (partial)
Cities covered	50+ currently, up to 100+ in 2 years
Service centres	400 now, scaling to 600+
Complaint volume	~50 per day currently, up to 200/day in 2 years
Daily active users	30–40 (staff + service centres)
Peak concurrent users	10–20
Total registered users	100–600 (service centres) + 2 admins

1.3 Platform Scope

- Web application accessible on laptop and desktop browsers.
- Progressive Web App (PWA) — installable on mobile and laptop home screen, no app store required.
- No native Android/iOS app required at this stage.
- WhatsApp Business API integration for 2 specific notification triggers only.

2. Roles & Access

2.1 Admin

- 2 admin accounts exist. Both see identical data and have identical permissions.
- Admin accounts are created manually — not through any self-registration flow.
- Admin is the only role that can register complaints, assign service centres, edit complaint details after SC submission, and view full billing across all centres.
- Admin can also manage presets and approve or reject service centre registrations.

2.2 Service Centre

- Each service centre represents a city-based repair/installation business.
- Each service centre is tagged with the product types they handle: LED only, Cooler only, or Both.
- Service centres self-register. Account is inactive until admin approves.
- After approval, they log in and see only complaints assigned to them.
- They cannot see other service centres' data or billing.
- They can accept or reject incoming assignments, update status, upload proof, fill customer payment amount (out of warranty), and manage petrol charges within allowed limits.

3. Authentication & Access Control

3.1 Login

- Single login page for all roles.
- Login via registered email and password.
- JWT-based session management.
- Role detected from token — UI adapts accordingly.

3.2 Service Centre Registration

Service centres register themselves through a public registration form. Fields required:

- Owner / contact person name
- Business name
- Full address (street, city, district, state)
- Phone number 1 (required), Phone number 2 (optional)
- Email 1 (required — used for login), Email 2 (optional — CC for notifications)
- Product capability: LED only / Cooler only / Both
- Password (set at registration)

After submission, the account is Pending. Admin sees it in the Action Centre and approves or rejects. On approval, the service centre receives an email and can log in. On rejection, they receive a rejection email.

3.3 Password Reset

- Available for all roles.
- User enters their registered email.
- System sends a 6-digit OTP to that email. OTP valid for 10 minutes.
- On correct OTP entry, user sets a new password.

4. Master Data

4.1 City & District Master List

A pre-seeded list of all cities and districts in Rajasthan and Punjab is stored in the database. Used across the system wherever city or district selection is needed.

- City selection always auto-fills district and state.
- No free-text city entry anywhere — all city fields are dropdowns from this master list.
- Admin can view the list (read-only in system).

4.2 Presets

Presets are standard charge packages used when registering a complaint. Each preset has:

- Type: installation_led / complaint_led / complaint_cooler
- Name (e.g. "Standard LED Installation", "Cooler Repair - Basic")
- Price (fixed amount in rupees)
- Status: active or inactive

Admin can create, edit, delete, or toggle active/inactive status of presets at any time. Only active presets appear in the complaint registration form. When a preset is selected for a complaint, the price is snapshotted and stored permanently — future edits to the preset do not affect existing complaints.

Presets apply only to in-warranty complaints. For out-of-warranty complaints, no preset is selected or charged from Microvison's side.

5. Products & Service Centre Capability

5.1 Product Types

Product	Allowed Complaint Types	Notes
LED	Installation OR Complaint	Only product that allows installation type
Cooler	Complaint only	No installation type available
Both (LED + Cooler)	Installation or Complaint for LED; Complaint only for Cooler	Used when a single visit covers both products

'Both' is a valid product selection on the complaint registration form. It means this single complaint/visit covers both LED and Cooler. The complaint type for each product is specified separately within the form.

5.2 Service Centre Product Capability

Each service centre declares what product types they can service at registration time:

- LED only — can only be assigned LED installation or LED complaint requests.
- Cooler only — can only be assigned Cooler complaint requests.
- Both — can be assigned any complaint type including combined LED + Cooler complaints.

This capability tag is used during complaint assignment:

- If a complaint is for LED only → system shows LED-only and Both-capable centres in the city.
- If a complaint is for Cooler only → system shows Cooler-only and Both-capable centres.
- If a complaint is for Both (LED + Cooler) → system shows only Both-capable centres.
- Admin sees this filtered list when assigning — capability mismatch centres are hidden or greyed out.

6. Complaint Registration

Only admin can register complaints. The form is divided into 4 steps.

6.1 Step 1 — Customer Information

Field	Type	Required	Notes
Customer name	Text	Yes	
Phone number 1	Number	Yes	Primary contact
Phone number 2	Number	No	Alternate contact
Local address	Free text	Yes	Street/locality/area
City	Dropdown	Yes	From master list
District	Auto-filled	Yes	Based on city selection
State	Auto-filled	Yes	Based on city selection

When admin enters Phone number 1, the system automatically checks if this customer has an open or recently closed complaint (within 30 days) for the same product and same complaint type. If found, a reopen option is shown as a banner. Admin can choose to reopen or continue creating a new complaint.

6.2 Step 2 — Product & Type

Admin selects one of three product options:

- LED — then selects type: Installation or Complaint
- Cooler — type is auto-set to Complaint (no installation option for coolers)
- Both (LED + Cooler) — admin selects type for LED (installation or complaint); Cooler type is auto-set to Complaint

Warranty selection (applies to the whole complaint):

- In Warranty — Microvison pays the service centre for the service rendered.
- Out of Warranty — customer pays the service centre directly. Microvison does not pay for the base service.

Warranty status is a single flag for the whole complaint, even if it covers both products.

6.3 Step 3 — Charges & Media

In Warranty — Preset selection

- Admin selects one active preset from the filtered list (filtered by product + type).
- For 'Both' complaints, admin selects one preset that best represents the combined service.
- Preset price is shown and locked permanently after complaint creation.
- Installation complaints use installation_led presets — price is fully fixed.
- Complaint type presets are a starting base — actual bill may include petrol and approved extras.

Out of Warranty — No preset

- No preset is shown or selected for out-of-warranty complaints.
- No petrol charge is applied from Microvison's side.
- Admin can optionally add extra charges (label + amount) if Microvison is contributing something specific (e.g. a spare part cost Microvison is covering).
- If no extras are added, Microvison generates no bill for this complaint — it is tracking only.
- If extras are added, a bill is generated for those extras only.

Petrol / diesel charge (in warranty only)

- Admin enters an estimated petrol/diesel amount at registration (optional but recommended).
- Editable up to 3 times total across all parties in this fixed order:
 - Edit 1 — Admin sets estimate at registration.
 - Edit 2 — Service centre adjusts to actual amount after the visit.
 - Edit 3 — Admin can override if the final amount seems incorrect.
- After the 3rd edit, the petrol field is permanently locked.
- All 3 values are stored separately for full audit trail.

Extra charges (both warranty types)

- Admin can add one or more extra charge line items at registration (label + amount).
- Service centre can also request additional charges after the visit.
- All SC-requested extra charges must be approved by admin before inclusion in the bill.
- Rejected extra charges are recorded but not included in billing.

Notes and media

- Text notes: free text, optional.
- Voice note: optional, max 60 seconds, uploaded as audio file.
- Photos: optional, maximum 2 images at registration stage.

6.4 Step 4 — Assign Service Centre

- System filters service centres by customer's city, then district.
- System further filters by product capability matching (see Section 5.2).
- Each matching centre is shown with current load stats (assigned, pending, done this month).
- Admin selects one service centre and submits.
- On submission: complaint is created, unique ID is generated, WhatsApp message is sent to the selected service centre.

7. Complaint ID & Status Lifecycle

7.1 Unique Complaint ID

MV-YYYY-XXXXX

- MV = Microvison prefix
- YYYY = year of creation
- XXXXX = auto-incremented 5-digit number, resets each year
- Reopened complaints get their own new ID, with a reference to the parent complaint ID stored internally.

7.2 Status Flow

Status	Set By	Meaning
new	System	Complaint created, not yet assigned
assigned	Admin	Admin has assigned a service centre
accepted	Service Centre	SC accepted the assignment
rejected_by_sc	Service Centre	SC rejected — admin must reassign
going	Service Centre	SC marked they are on the way (optional, skippable)
done	Service Centre	Work completed successfully
not_done	Service Centre	Visit happened but work could not be completed
part_pending	Service Centre	Part unavailable — follow-up needed
replacement	Service Centre	Product needs replacement — escalation required
reopened	Admin	Same issue recurred within 30 days — complaint reopened
closed	Admin	Admin confirmed done — complaint locked and bill generated

Bill is generated only when admin confirms done (status = closed). No bill is generated for not_done, part_pending, or replacement until the issue is fully resolved.

7.3 Status Transition Rules

- new → assigned: admin only.
- assigned → accepted / rejected_by_sc: assigned service centre only.
- rejected_by_sc → assigned: admin reassigns to a different centre.

- accepted → going: optional, skippable, service centre only.
- accepted / going → done / not_done / part_pending / replacement: service centre, after uploading proof photos.
- done → closed: admin reviews, optionally edits, then confirms. Bill is generated at this point.
- done / not_done → reopened: only if reopen eligibility conditions are met (see Section 8).

8. Complaint Reopen Logic

A complaint can be reopened when all of the following are true:

- Same customer (matched by phone number 1).
- Same product (LED / Cooler / Both).
- Same complaint type (installation / complaint).
- Original complaint was created within the last 30 days.
- Original complaint status is done or not_done.

If the customer calls about a different problem on the same product, a new complaint is always created. There is no automatic reopen — admin makes the decision from the banner shown during complaint registration.

When a complaint is reopened:

- A new complaint record is created with a new MV-YYYY-XXXXX ID.
- The new record stores a reference to the original complaint ID.
- Admin adds reopen notes (text, required) and optional photos.
- A WhatsApp message is sent to the service centre with REOPENED label.
- The reopened complaint goes through the same full workflow from assigned onwards.
- Billing for the reopened complaint is separate from the original.

9. Admin Edit After Service Centre Submission

After a service centre fills in the complaint result (status, proof photos, petrol actual, extra charge requests, notes), the complaint appears in the admin Action Centre for review.

Before confirming done and generating a bill, admin has the ability to edit the following fields:

- Petrol amount (if edit count allows — i.e. it is admin's turn, edit #3).
- Approve or reject extra charge requests submitted by the SC.
- Add admin-side extra charges if needed.
- Add or update admin notes on the complaint.
- Revert status back to accepted (if admin disputes the done marking) with a note to the SC.

After admin is satisfied with all details, they confirm done. This action:

- Locks the complaint status as closed.
- Generates and locks the bill.
- No further edits are possible after this point.

10. Service Centre Portal

After login, the service centre sees their dashboard with 3 tabs.

10.1 Tab 1 — New Requests

Lists all complaints assigned to this service centre that have not yet been accepted or rejected.

- Each card shows: complaint ID, customer local address, city, product, complaint type, warranty status, preset name and price (in-warranty only), admin notes, voice note player, admin photos.
- Two actions: Accept or Reject.
- On Accept: status moves to accepted.
- On Reject: status moves to rejected_by_sc. Goes back to admin for reassignment.

10.2 Tab 2 — My Complaints

Lists all complaints assigned to this service centre across all statuses.

- Filters: status, product type, complaint type, warranty status, date range.
- Ordered by most recently assigned first.

Clicking a complaint opens the detail view. Actions available:

- Mark as Going (optional, skippable) — when status is accepted.
- Upload proof photos (0 to 5 images, mandatory before marking final status). Includes warranty slips, work-done evidence, site photos.
- Mark final status: Done / Not Done / Part Pending / Replacement.
- Edit petrol actual amount — available only if it is SC's turn (edit #2, in-warranty complaints only).
- Request extra charges — label and amount, sent to admin for approval.
- Add extra notes (text, optional).

Out of warranty — customer payment field

- For out-of-warranty complaints, after marking status (done / not done), a single field appears:
 - "Amount collected from customer" — filled by service centre, single number field.
 - This is the total cash or UPI amount the customer paid the service centre directly.
 - This field is for Microvison's record only — it does not appear in any Microvison invoice.
 - This field is required before the SC can submit their final status update on out-of-warranty complaints.

10.3 Tab 3 — Billing

- Lists all complaints that have a bill generated for this service centre (in-warranty, closed complaints with preset or approved extras).
- Per-complaint breakdown: preset amount, petrol final, approved extras, total.
- Monthly invoice view: auto-calculated sum of all complaint bills in a given month.
- Filter by month and year.

Out-of-warranty complaints with no admin extras do not appear in the billing tab — they have no bill from Microvison's side. Out-of-warranty complaints where admin added extras do appear, showing only the extra charges billed.

11. Admin Portal

Admin dashboard has 4 main tabs plus a settings section for presets.

11.1 Tab 1 — Action Centre

The admin's notification and task hub. Shows items requiring attention, ordered newest first. Badge count on tab for pending items.

Items shown in Action Centre

- Pending service centre registration requests — Approve or Reject.
- Complaints marked Done by SC — admin reviews, optionally edits, then confirms (generates bill) or disputes (reverts to accepted).
- Complaints marked Not Done / Part Pending / Replacement — shown for awareness and follow-up.
- Complaints rejected by a service centre — admin must reassign.
- Extra charge requests from service centres — Approve or Reject each line item.

11.2 Tab 2 — Service Centres

Full list of all registered service centres (approved, pending, and rejected).

Search and filter options:

- Search by business name, owner name, city, district, phone number.
- Filter by status: active, pending, rejected.
- Filter by product capability: LED only, Cooler only, Both.

Each card shows: name, city, phone, product capability, live stats (total assigned, currently pending, completed this month, rejected count).

Clicking a service centre opens detail view showing full registration info (editable by admin), all complaints assigned to them, and option to deactivate.

11.3 Tab 3 — All Complaints

Complete list of every complaint in the system, ordered by creation date (newest first).

Search options:

- Customer name, customer phone number (1 or 2), complaint ID.

Filter options:

- City, district, state
- Product type (LED, Cooler, Both)

- Complaint type (installation, complaint)
- Warranty status (in warranty, out of warranty)
- Status (any single or multiple statuses)
- Assigned service centre
- Service centre product capability
- Date range (created between X and Y)
- Reopened complaints only (toggle)

Each row: complaint ID, customer name, city, product, type, warranty status, current status, assigned centre, creation date.

Clicking opens the full complaint detail view. Admin actions from this view:

- Reassign to a different service centre (when status is new, assigned, or rejected_by_sc).
- Edit petrol, approve/reject extras, add admin notes (when status is done, before confirming).
- Confirm done → generates and locks bill.
- Dispute done → reverts to accepted with a note to SC.
- Initiate reopen (if eligibility conditions are met).

11.4 Tab 4 — Billing

Billing view across all service centres and all complaints.

Filters: same as complaints tab, plus month/year selection and service centre name.

Two views:

- Per-complaint billing: each complaint with full charge breakdown (preset + petrol + approved extras = total). Out-of-warranty complaints show only extras if any; also shows customer payment amount (what SC collected) as a separate read-only field.
- Monthly invoice per centre: auto-calculated rollup of all Microvison-side bills for a centre in a given month. Shows number of complaints and total amount Microvison owes the centre.

The customer payment amount (out-of-warranty) is visible in the billing view for record purposes but is never included in Microvison's invoice to the service centre.

11.5 Presets Management

- View all presets grouped by type.
- Create new preset — type, name, price.
- Edit existing preset — name or price (does not affect already-created complaints).
- Delete preset — only allowed if no complaints are currently using it.
- Toggle active/inactive — inactive presets do not appear in complaint registration form.

12. Complaint Detail View

Shared by admin and service centre. Available actions differ by role.

Section	Fields Displayed
Complaint header	Complaint ID, creation date, current status, reopened flag, parent complaint ID (if reopen)
Customer	Name, Phone 1, Phone 2, local address, city, district, state
Product	Product selection (LED/Cooler/Both), complaint type, warranty status
Preset & base charges	Preset name and price — shown only for in-warranty complaints. Locked after creation.
Petrol charges	Admin estimate, SC actual, Admin override, edit count, locked flag. In-warranty only.
Extra charges	All line items — label, amount, requested by, status (pending / approved / rejected)
Notes & media	Admin text notes, voice note player, admin registration photos
SC proof media	Proof photos uploaded by SC after visit
Customer payment	Amount collected from customer — shown for out-of-warranty complaints only. Read-only for admin.
Reopen details	Reopen notes and photos (if this complaint is a reopen)
Status timeline	Every status change with timestamp, who changed it, and any note at time of change
Bill summary	Shown only when status is closed — full breakdown and total. For out-of-warranty, shows extras only (if any).

13. Billing Logic

13.1 In Warranty — Bill Components

Component	Source	Editable After Complaint Creation?
Preset base price	Selected preset, snapshotted at creation	No — locked permanently
Petrol / diesel	3-round edit process (admin → SC → admin)	Locked after 3rd edit
Approved extra charges	SC requests approved by admin, or admin-added	No — locked after approval
Total	Preset + petrol + approved extras	Auto-calculated

13.2 Out of Warranty — Bill Components

Component	Source	Notes
Preset	None	Not applicable — Microvison does not pay base service charge
Petrol	None	Not applicable — Microvison does not cover petrol for out-of-warranty
Admin extras (optional)	Admin adds if Microvison is contributing a specific item	If added, a bill is generated for these extras only
Total (Microvison side)	Admin extras only (if any)	Zero if no extras added
Customer payment (record only)	Filled by service centre — what customer paid them directly	Never included in Microvison invoice. For records only.

13.3 When Bills Are Generated

Scenario	Bill Generated?
In warranty — admin confirms done (status = closed)	Yes — full bill (preset + petrol + approved extras)
Out of warranty — no admin extras — admin confirms done	No bill — tracking record only
Out of warranty — admin extras present — admin confirms done	Yes — bill for extras only
Status is not_done	No
Status is part_pending or replacement	No — bill generated only after fully resolved and closed
Reopened complaint — admin confirms done	Yes — separate bill, separate entry, references original

13.4 Monthly Invoice

- Auto-calculated — sum of all Microvison-side complaint bills for a given service centre in a calendar month.
- Always a live calculation — no separate generation step.
- Admin can view invoices for any centre, any month.
- Service centre can view only their own invoices.
- Customer payment amounts are visible in the admin billing view for reference but are excluded from the monthly invoice total.

14. WhatsApp Notifications

WhatsApp Business API is used for 2 notification triggers only. All other updates are in-system.

14.1 Trigger 1 — New Complaint Assigned

Sent to the service centre's registered phone number immediately after admin assigns the complaint.

Message content

- Label: NEW COMPLAINT ASSIGNED
- Complaint ID
- Customer name and Phone 1
- Customer local address, city
- Product and complaint type
- Warranty status
- Preset name and price (in-warranty only)
- Admin petrol estimate (if entered, in-warranty only)
- Admin notes (if any)
- Portal login URL

14.2 Trigger 2 — Complaint Reopened

Sent to the service centre assigned to the reopened complaint, immediately after admin creates the reopen.

- Label changed to: COMPLAINT REOPENED
- Reference to original complaint ID shown.
- Admin reopen notes included.
- All other fields same as Trigger 1.

WhatsApp provider (Twilio, WATI, or AiSensy) to be finalized separately. Message content remains the same across all providers — only API credentials change.

15. Progressive Web App (PWA)

- The web application must be installable as a PWA on both mobile phones and laptops.
- On Android/Chrome: users see an 'Add to Home Screen' prompt. App opens with the Microvision icon, no browser chrome visible.
- On laptop/Chrome: users see an install icon in the address bar.
- No separate app store submission or native code required.
- When the main website is updated and redeployed, the PWA auto-updates silently in the background on the user's next visit.
- Requires: manifest.json (app name, icons, theme colour) and a service worker for caching and update handling.
- Offline behaviour: app shell (login page, dashboard layout) loads offline. Data-dependent screens require internet connection.

16. Edge Cases & Business Rules

Scenario	Behaviour
SC rejects a complaint	Status → rejected_by_sc. Appears in admin Action Centre. Admin must reassign to a different centre.
Admin tries to assign to same SC that rejected	System warns but does not block. Admin override allowed.
Complaint marked done, admin disputes	Admin reverts status back to accepted. SC is expected to re-update. Note added to timeline.
Same customer, different product	Always a new complaint — reopen logic does not apply across different products.
Same product, different complaint type	Always a new complaint (e.g. LED installation is separate from LED complaint).
Preset deleted after complaint creation	Preset price is snapshotted — deletion does not affect existing complaints.
SC deactivated mid-complaint	Existing assigned complaints are not removed. Admin should manually reassign.
Petrol field left blank (in-warranty)	Valid — bill generated without petrol. Total = preset + approved extras only.
Extra charge rejected by admin	Stored as rejected — visible in complaint history but not included in bill.
Bill generated, then extra charge approved	Bill is recalculated to include the newly approved charge.
Out of warranty, no extras, SC fills customer payment	Complaint closes as tracking record only. No Microvison bill generated.
Out of warranty, SC does not fill customer payment field	System blocks final status submission until the field is filled.
Both product complaint assigned to LED-only SC	System prevents this — only Both-capable centres are shown for Both complaints.
Voice note exceeds 60 seconds	System rejects the file and shows an error. User must re-record.
SC uploads more than 5 proof photos	System allows max 5. Error shown if 6th is attempted.

17. Data Models Summary

users

Field	Type	Notes
_id	ObjectId	Auto-generated
name	String	Full name
email	String	Unique, used for login
passwordHash	String	Bcrypt hashed
role	Enum	admin / service_centre
status	Enum	active / pending / rejected
createdAt	Date	

serviceCentres

Field	Type	Notes
_id	ObjectId	
userId	Ref: users	Linked login account
ownerName	String	
businessName	String	
phone1	String	Primary
phone2	String	Optional
email1	String	Primary / login email
email2	String	Optional CC
fullAddress	String	Street/locality
city	String	From master list
district	String	Auto from city
state	String	Auto from city
productCapability	Enum	led_only / cooler_only / both
status	Enum	active / pending / rejected
createdAt	Date	

complaints

Field	Type	Notes
_id	ObjectId	
complaintId	String	MV-YYYY-XXXXX
customerName	String	

Field	Type	Notes
phone1	String	
phone2	String	Optional
localAddress	String	
city / district / state	String	From master list
product	Enum	led / cooler / both
complaintType	Enum	installation / complaint
warrantyStatus	Enum	in_warranty / out_of_warranty
presetId	Ref: presets	Null for out-of-warranty
presetPrice	Number	Snapshoted at creation. Null for out-of-warranty.
petrolAdmin	Number	Edit 1. In-warranty only.
petrolSC	Number	Edit 2. In-warranty only.
petrolFinal	Number	Edit 3 admin override. In-warranty only.
petrolEditCount	Number	0–3. In-warranty only.
petrolLocked	Boolean	True after 3rd edit.
extraCharges	Array	[[label, amount, requestedBy, status, approvedAt]]
customerPaymentAmount	Number	Out-of-warranty only. Filled by SC. Record only.
notes	String	Admin text notes
voiceNoteUrl	String	Cloudinary URL
adminPhotos	Array	Max 2 Cloudinary URLs
proofPhotos	Array	Max 5 Cloudinary URLs — SC uploads
assignedCentredId	Ref: serviceCentres	
assignedAt	Date	
status	Enum	See Section 7.2
isReopened	Boolean	
reopenedAt	Date	
reopenParentId	Ref: complaints	If this is a reopen
reopenNotes	String	Required on reopen
reopenPhotos	Array	Optional
billGenerated	Boolean	
billLockedAt	Date	
whatsappSent	Boolean	
createdBy	Ref: users	Admin who created
createdAt / updatedAt	Date	

Other collections

Collection	Purpose
cities	Master list of cities with district and state — seeded once for Rajasthan + Punjab
presets	Charge packages — type, name, price, active status
complaintUpdates	Audit log — every status change with timestamp, role, user, note
invoices	Monthly invoice records per service centre — auto-calculated, month+year keyed
otpTokens	Short-lived OTP records for password reset — email, code, expiry

18. Recommended Build Order

Phase	Items
1 — Foundation	DB setup, all schemas, city master data seed (Rajasthan + Punjab)
2 — Auth	Login, JWT middleware, RBAC, SC registration with capability field, admin approval, OTP password reset
3 — Presets	Admin preset CRUD — create, edit, delete, toggle active
4 — Complaint registration	Full 4-step form, duplicate/reopen check, product capability filtering in SC assignment, WhatsApp trigger on submit
5 — SC portal	New requests tab, accept/reject, my complaints tab, status updates, proof upload, petrol edit, extra charge requests, customer payment field for out-of-warranty
6 — Admin action centre	SC registration approvals, done confirmations, extra charge approvals, reassignment handling
7 — Admin complaints tab	Full list, all search and filter options including capability filter, complaint detail view with admin edit and confirm actions
8 — Billing	Bill generation logic, in-warranty vs out-of-warranty split, per-complaint breakdown, monthly invoice calculation, billing tabs for both roles
9 — Admin service centres tab	SC list with capability filter, search, stats, detail view, deactivate
10 — Reopen flow	Reopen detection on phone entry, reopen form, second WhatsApp trigger
11 — PWA	manifest.json, service worker, offline shell, auto-update on deploy
12 — WhatsApp provider	Connect chosen provider (Twilio / WATI / AiSensy), test both triggers
13 — QA & deploy	End-to-end testing, Railway backend deploy, Vercel frontend deploy, domain and email config

19. Out of Scope

- AC Stabilizers — not a product line in this system.
- Native Android or iOS app — PWA covers mobile needs.
- Customer-facing portal — complaints are registered and managed internally by admin only.
- Real-time push notifications via WebSocket — Action Centre polling is sufficient at this scale.
- Payment gateway integration — billing is tracked internally, no online payment collection.
- SMS notifications — WhatsApp covers communication needs.
- Multi-language support.
- Reporting or analytics dashboards beyond the billing and complaints tabs described.
- Integration with external CRM or ERP systems.
- Automated complaint assignment — admin always assigns manually.

End of Document · Microvison GRD v1.1